Fleet condition report: Synthetic ACME-style vendor export generated from Kelmarsh KEL01-02

Synthetic ACME-style vendor export generated from Kelmarsh KEL01-02, 2022. 2 turbines, SCADA 01 Jan 2022 to 31 Dec 2022; assessment window: the last 24 months of each turbine (Jan 2022 to Dec 2022). Generated by `windml report` from the mapping `acme_messy.yaml`.

Executive summary

No turbine shows a warning sign in the last 24 months of data.

No action needed now: KEL01, KEL02.

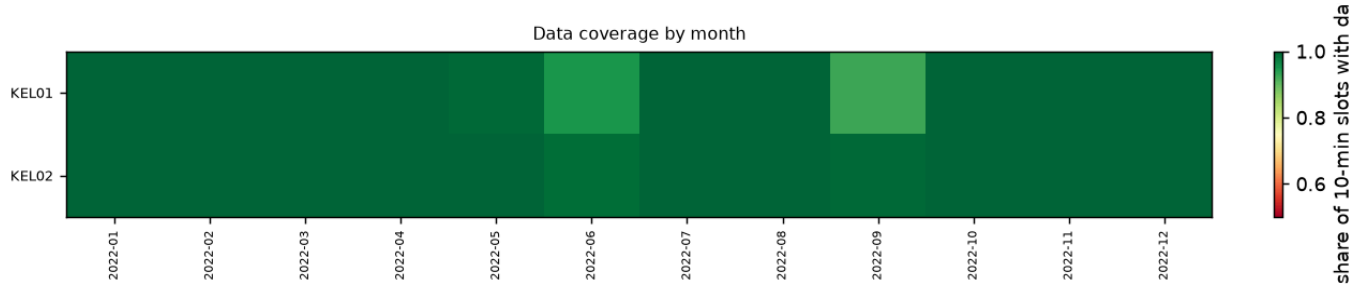
How sure are we. Confidence is *High* when the turbine has 3+ years of history, near-complete data, models trained on earlier years and an alarm log; it drops when any of these is missing. A red flag is a reason to look, not a diagnosis.

What this report could not check:

- The history is short (under two years), so models were cross-fitted within the record: they spot sudden changes and outliers, but a slow change that started before the data begins is invisible.
- The export has no pitch-motor temperatures, so those components could not be checked.

1. Data quality

Turbine	First data	Last data	Months	Coverage (window)	Longest gap (h)	Signals	Native interval	Frozen rows (window)	Stuck sensors (>= 1%)	Stop/fault labels
KEL01	2022-01-01	2022-12-31	12	99.0%	49	6	10min	0	-	alarm log
KEL02	2022-01-01	2022-12-31	12	99.9%	5	6	10min	0	-	alarm log



Signals present in at least one turbine: `ambient_temp_c`, `gear_oil_inlet_pressure_bar`, `gear_oil_temp_c`, `generator_speed_rpm`, `main_bearing_front_temp_c`, `nacelle_position_deg`, `nacelle_temp_c`, `pitch_a_deg`, `power_kw`, `rotor_speed_rpm`, `wind_direction_deg`, `wind_speed_ms`.

2. Turbine scorecard

Turbine	Flag	Confidence	Tech. avail.	Energy avail.	Fault h/yr	Largest fault cause	Oil ep.	Pitch ep.	Abnormal oil days, final yr	Under-perf. days, final yr	Pump pressure: base to last (bar)	Oil-temp model error
KEL01	GREEN	Medium	98.5%	-	17	hydraulics (8 h)	0	1	2%	0%	-	-
KEL02	GREEN	Medium	97.3%	-	12	converter (6 h)	1	0	2%	0%	-	-

Flags and reasons

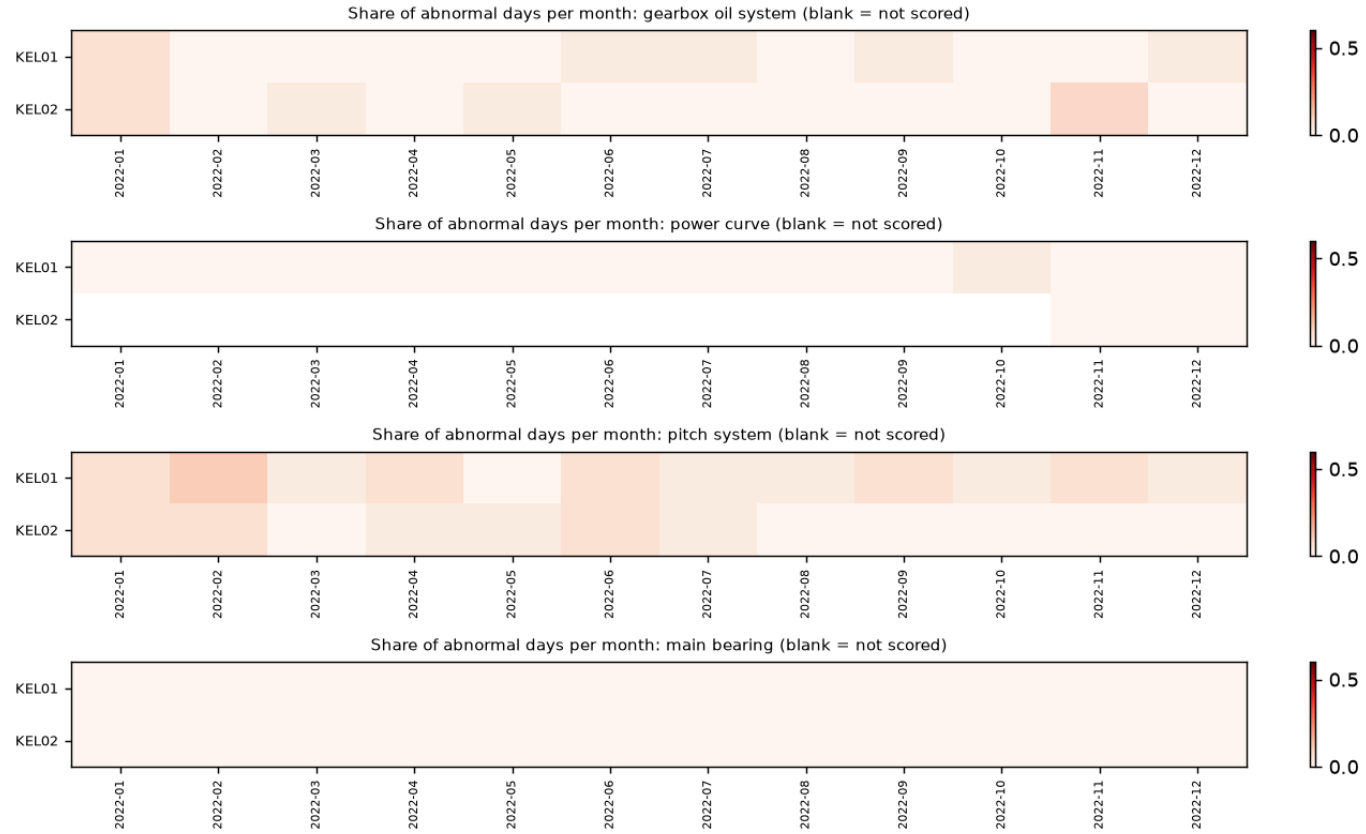
- **KEL01** GREEN (confidence Medium: 12 months of history, short-history models (cross-fit)). No trigger.
- **KEL02** GREEN (confidence Medium: 12 months of history, short-history models (cross-fit)). No trigger.

3. Model-based health (normal-behaviour residuals)

Each turbine gets its own models that predict a signal (oil temperature, bearing temperature, pitch angle, power...) from operating conditions. A day is *abnormal* when the turbine's median deviation from its own model is beyond 3 robust standard deviations in the harmful direction. *Chronic* means at least 30% abnormal days in a month.

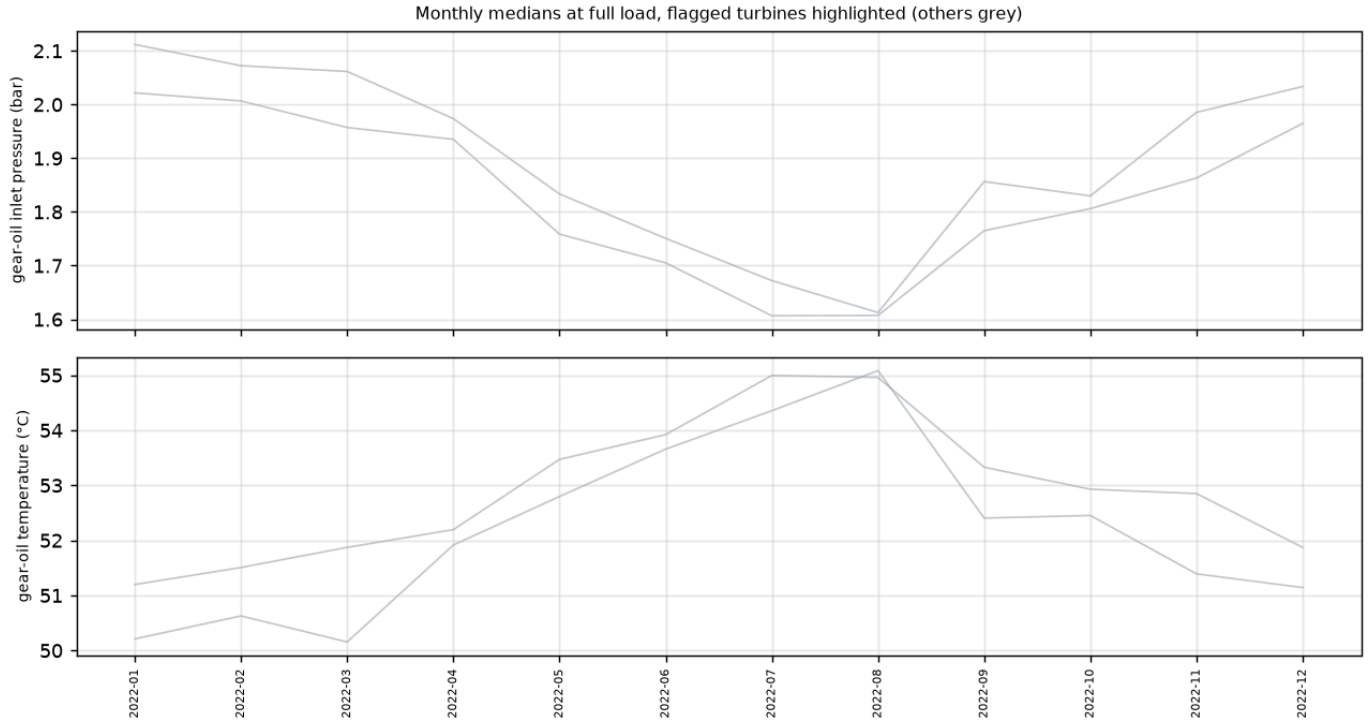
Turbine	Training scheme	Models	Window scored	gearbox oil system: abnormal days final yr / chronic months	main bearing: abnormal days final yr / chronic months	pitch system: abnormal days final yr / chronic months	power curve: abnormal days final yr / chronic months
KEL01	cross-fit	5	99%	2% / 0	0% / 0	5% / 0	0% / 0
KEL02	cross-fit	5	97%	2% / 0	0% / 0	3% / 0	0% / 0

No turbine has 3 or more consecutive chronic months in the window.



4. Gearbox oil pressure and temperature trends

Turbine	Pump pressure (bar): baseline → last month	Inlet pressure (bar)	Oil temperature (°C)	Baseline months
KEL01	-	2.0 → 2.0 (+0.0)	51.3 → 51.1 (-0.1)	2022-01..2022-06
KEL02	-	1.9 → 2.0 (+0.0)	52.0 → 51.9 (-0.2)	2022-01..2022-06

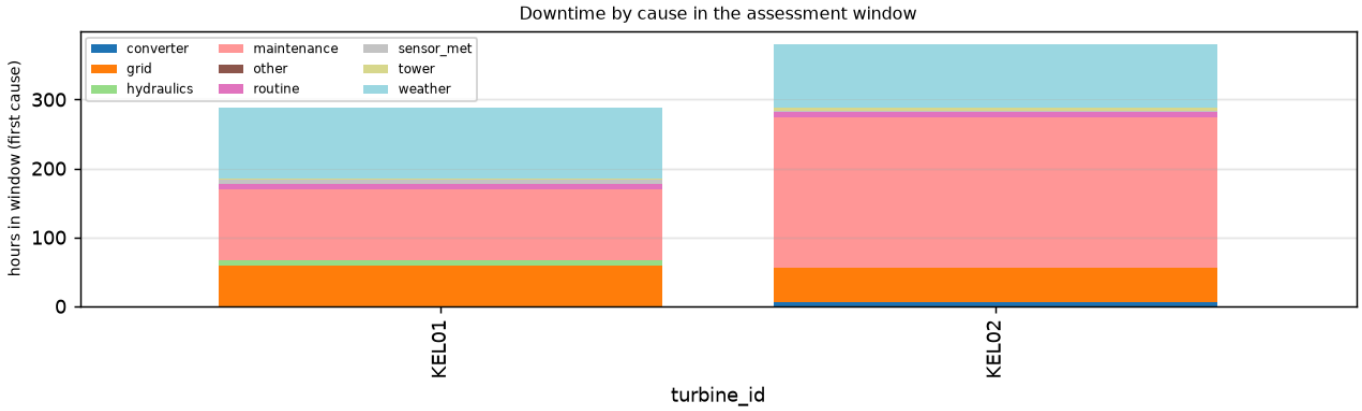


5. Sensor drift and stuck sensors

Fewer than 4 turbines share a drift-check signal, so sensor drift against the fleet could not be checked.

6. Stop and fault history

Turbine	Fault stops	Maintenance	Grid	Weather	Curtailment	Major-fault episodes	Maint. h/yr	Oil / pitch trips
KEL01	75	36	4	12	0	1	102	0 / 1
KEL02	35	30	18	3	0	1	218	1 / 0



Most frequent fault stops in the window (all turbines):

Code	Message	Component	Stops
6530	Anemometer defect	sensor_met	70
4540	Tower oscillation X level 2	tower	12
1070	Drive train monitor level 2	gearbox	6
4510	Tower oscillation Y level 1	tower	6
6620	Vane defect	sensor_met	4
3110	Frequency converter error	converter	4
5510	Low hydraulic pressure	hydraulics	2
4520	Tower oscillation X level 1	tower	2
1065	Drive train monitor level 1	gearbox	1
815	Rotor overspeed nacelle	rotor	1
455	Repeating error BP52	pitch	1
1800	Overload gear oil pump	gearbox	1

7. Methods, thresholds and limits

- **Pipeline.** `windml report` reads the raw export only through the mapping (`windml.ingest`), cleans and labels each turbine (QC flags, short-gap interpolation, alarm-log stop labels), trains the models, computes the metrics and writes this report. Every number here is reproducible from the mapping and the raw files.
- **Window.** The last 24 months of each turbine; *final year* = its last 12 months.
- **Models.** LightGBM per turbine and signal on healthy operating rows (no stop within 6 h, no fault, not in the 30 days before a major fault, QC clean). With 2+ calendar years, each year is scored by a model trained only on earlier years (3-day embargo, training must cover all 12 months). With 12-23 months, the record is cut into 6 blocks scored by models trained on the other blocks (cross-fit). Under 12 months, no model.
- **RAG rules** (fixed in advance, heuristics, not a calibrated risk): RED for availability < 95%, >= 8 oil or pitch trip episodes, pump pressure >= 2 bar above baseline, or abnormal oil state on >= 30% of final-year days; AMBER for availability < 97%, >= 4 episodes, pump rise >= 1 bar, abnormal oil days >= 10%, oil-temperature model error >= 1.5x its pre-window level, or > 2% missing data; AMBER only for under-performance on >= 20% of final-year days, abnormal main-bearing or pitch residuals on >= 30%, sensor drift (1.5 °C, 5% wind) or a stuck monitored sensor.
- **Confidence.** Points for history (3+ years: 2, 2-3 years: 1), coverage >= 95%, models trained on earlier years, an alarm log, and (for flags) agreement of independent indicators. High >= 4, Medium 2-3, Low <= 1.
- **Limits.** The alarm-log classes come from the mapping's rules; trip detection uses Senvion-style oil/pitch codes and messages (other vendors need `class_rules`). Full-load oil medians assume one oil circuit design across the fleet. Models learn whatever was normal in their training data, including a fault that was already present. Sensor drift is relative to the other turbines, so a fleet-wide drift is invisible.
- **Data.** Derived from Kelmarsh SCADA, CC BY 4.0, Cubico Sustainable Investments, doi:10.5281/zenodo.16807551.